

Eléments de trigonométrie

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	$\times$

$$\begin{aligned}
 \cos\left(\frac{\pi}{2} - \theta\right) &= \sin \theta & \sin\left(\frac{\pi}{2} - \theta\right) &= \cos \theta \\
 \cos\left(\frac{\pi}{2} + \theta\right) &= -\sin \theta & \sin\left(\frac{\pi}{2} + \theta\right) &= \cos \theta \\
 \cos(\pi - \theta) &= -\cos \theta & \sin(\pi - \theta) &= \sin \theta \\
 \cos(\pi + \theta) &= -\cos \theta & \sin(\pi + \theta) &= -\sin \theta
 \end{aligned}$$

$$\begin{aligned}
 \sin^2(x) + \cos^2(x) &= 1 \\
 1 + \tan^2(x) &= \frac{1}{\cos^2(x)}, \quad x \notin \frac{\pi}{2} + \pi\mathbb{Z}.
 \end{aligned}$$

$$\begin{aligned}
 \cos(a + b) &= \cos a \cos b - \sin a \sin b \\
 \cos(a - b) &= \cos a \cos b + \sin a \sin b \\
 \sin(a + b) &= \sin a \cos b + \cos a \sin b \\
 \sin(a - b) &= \sin a \cos b - \cos a \sin b \\
 \tan(a + b) &= \frac{\tan a + \tan b}{1 - \tan a \tan b}, \quad a, b, a + b \notin \frac{\pi}{2} + \pi\mathbb{Z}. \\
 \tan(a - b) &= \frac{\tan a - \tan b}{1 + \tan a \tan b}, \quad a, b, a - b \notin \frac{\pi}{2} + \pi\mathbb{Z}.
 \end{aligned}$$

$$\begin{aligned}
 \sin(2x) &= 2 \sin(x) \cos(x) \\
 \cos(2x) &= \cos^2(x) - \sin^2(x) = 2 \cos^2(x) - 1 = 1 - 2 \sin^2(x) \\
 \tan(2x) &= \frac{2 \tan(x)}{1 - \tan^2(x)}, \quad x \notin \pi\mathbb{Z}.
 \end{aligned}$$

$$\begin{aligned}
 t = \tan\left(\frac{\theta}{2}\right) \quad \cos(\theta) &= \frac{1 - t^2}{1 + t^2} \quad \sin(\theta) = \frac{2t}{1 + t^2}, \quad \theta \notin \pi\mathbb{Z}, \\
 \tan(\theta) &= \frac{2t}{1 - t^2}, \quad \frac{\theta}{2}, \theta \notin \frac{\pi}{2}\mathbb{Z}.
 \end{aligned}$$

DL de référence

$$e^x = 1 + x + \frac{1}{2}x^2 + \cdots + \frac{1}{n!}x^n + o(x^n)$$

$$\cos(x) = 1 - \frac{1}{2}x^2 + \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + o(x^{2n+1})$$

$$\sin(x) = x - \frac{1}{6}x^3 + \cdots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + o(x^{2n+2})$$

$$\arctan(x) = x - \frac{1}{3}x^3 + \cdots + \frac{(-1)^n}{2n+1}x^{2n+1} + o(x^{2n+2})$$

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + o(x^n)$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots + \frac{(-1)^{n-1}}{n}x^n + o(x^n)$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + o(x^n)$$